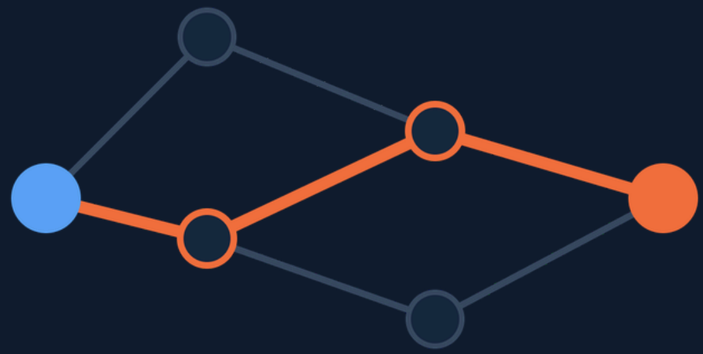
ARTIFICIAL INTELLIGENCE · GROUP PRESENTATION

AI-Based Ambulance Route Optimization



Finding the fastest, safest route to the patient and on to the right hospital, using A* search.



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PROBLEM AND DOMAIN

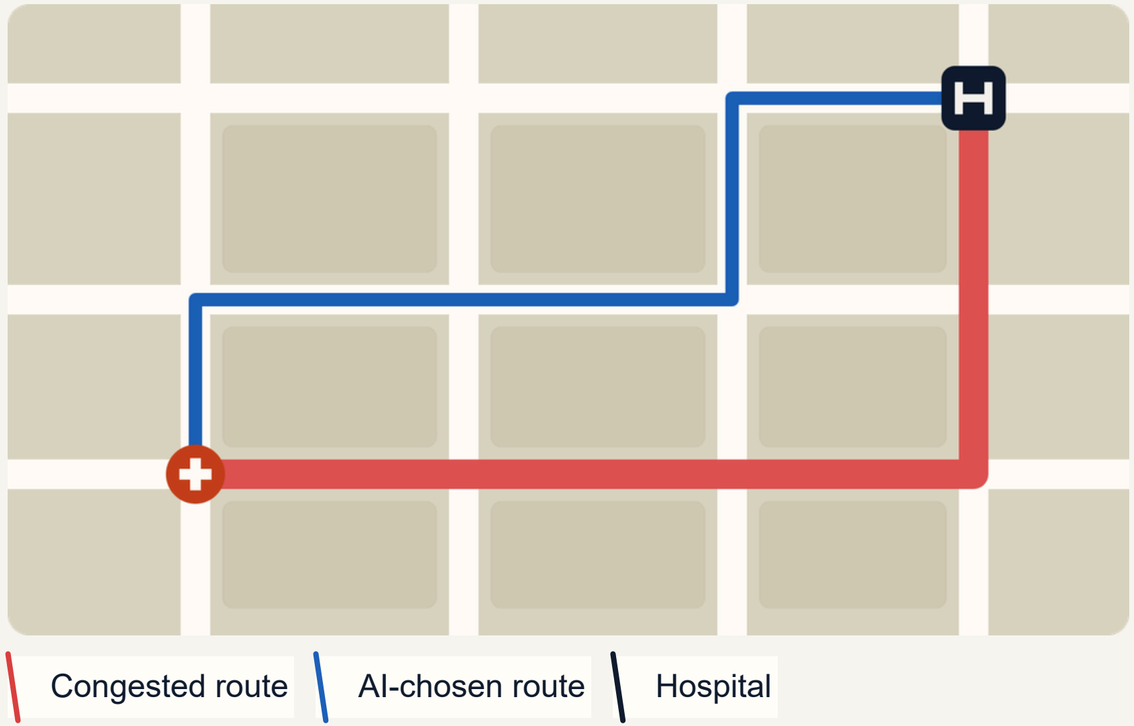
Delayed ambulances cost lives

Application domain
Emergency medical services, smart-city traffic

The problem
Congestion, signals and poor hospital choice

Why it matters
Survival falls with every minute of delay

Goal: shorten the time from emergency call to treatment.



PROBLEM AND DOMAIN

Why current routing falls short

Current approach	What goes wrong
Static shortest-distance maps	Shortest distance is not the fastest path in a jam.
Delayed traffic updates	The route is stale before the next junction.
Signals and crossings ignored	Red lights and crossings add uncontrolled delay.
Nearest-hospital-only choice	The nearest hospital may lack beds or a specialist.

AI closes the gap: predict, search and re-plan in real time.

PROPOSED SOLUTION

Predict, search, re-plan

Emergency call

Caller location and patient condition are captured at dispatch.

Data fusion

GPS, live traffic, signal status and hospital bed data are combined.

Travel-time prediction

A machine-learning model estimates the driving time of each road segment.

Route search

A* searches the road graph for the lowest-time route to the best hospital.

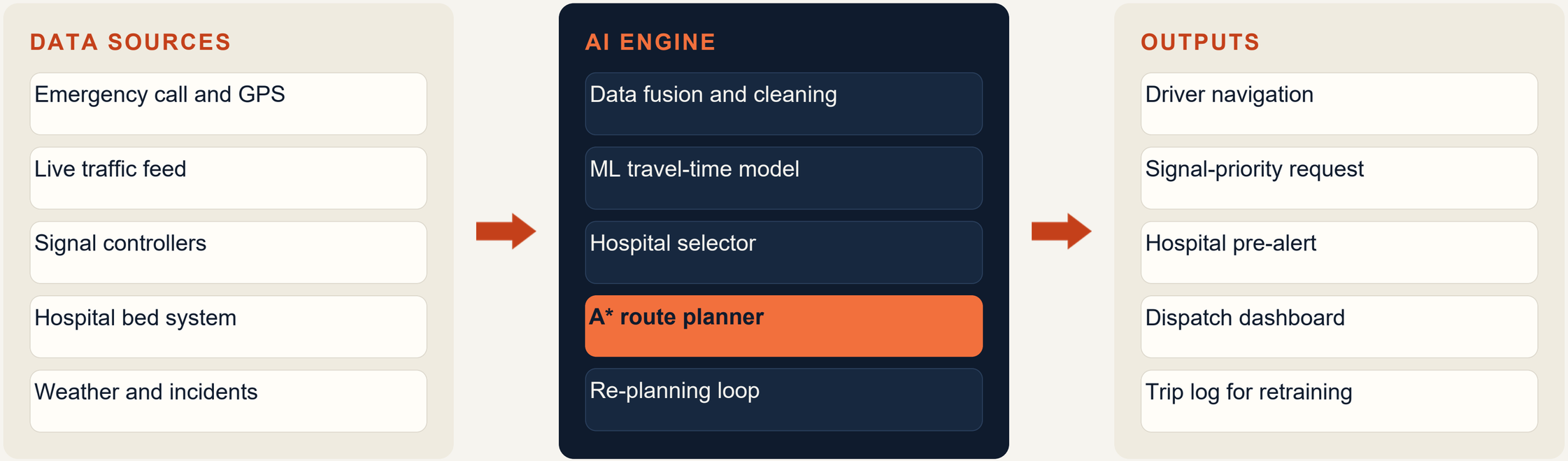
Guidance and re-planning

Turn-by-turn guidance, signal-priority requests and live re-routing.

Output: the fastest safe route, refreshed every few seconds as the ambulance moves.

PROPOSED SOLUTION

System architecture



Feedback loop: completed trips are logged and used to retrain the travel-time model.

FUNDAMENTAL AI CONCEPTS

AI concepts behind the solution

Intelligent agent

Senses traffic and hospital data, then acts by choosing and updating a route.

State-space model

Junctions are states and road segments are actions on a graph.

Heuristic search

Uses the straight-line time to the hospital to focus the search.

Machine learning

Predicts road travel times from historical and live traffic data.

Optimization

Minimizes total travel time while meeting hospital constraints.

Planning under uncertainty

Re-plans when traffic changes, because the environment is dynamic.

Modelling the agent (PEAS)

PEAS element	Ambulance routing agent
Performance	Minimum travel time, safe path, suitable hospital
Environment	Road network, live traffic, signals, weather, hospitals
Actuators	Route display, signal-priority requests, hospital alerts
Sensors	GPS, traffic feeds, road cameras, hospital bed data

Environment type: partially observable, dynamic, stochastic and sequential.

SEARCH STRATEGY

Formulating the search problem

STATE

Current junction and time of day

INITIAL STATE

Junction nearest the emergency

ACTIONS

Drive along a road leaving the junction

TRANSITION

Arrive at the junction at the road's far end

GOAL TEST

Entrance of a suitable hospital

PATH COST

Predicted travel time of the road (min)

A solution is a sequence of junctions from the accident site to a hospital.

SEARCH STRATEGY

A* search: f(n) = g(n) + h(n)

g(n)

Cost so far

Predicted travel time from the accident site to junction n.

h(n)

Heuristic

Straight-line distance to the hospital divided by top ambulance speed.

f(n)

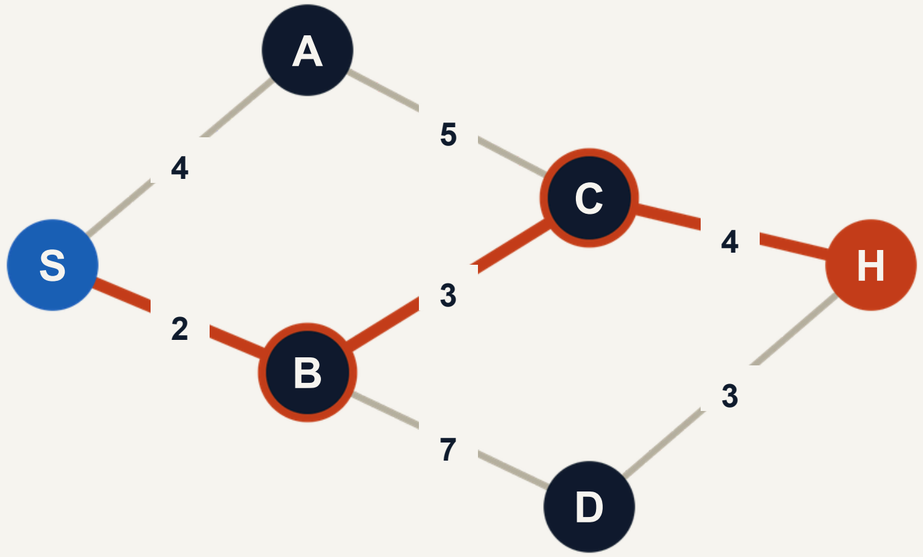
Priority

Estimated total trip time through n. A* always expands the lowest f first.

h(n) never overestimates, so the route A* returns is the fastest one.

APPLYING THE STRATEGY

Worked example: A* on a toy network



Step	Expand	Frontier (g+h=f)
1	S	B 2+6=8 · A 4+6=10
2	B	C 5+3=8 · A 10 · D 12
3	C	H 9+0=9 · A 10 · D 12
4	H	Goal reached: 9 min

Best route: S, B, C, H in 9 minutes. A* expands only S, B, C and H; A and D are never expanded.

Edge labels are predicted minutes. Heuristic h (min): S 8 · A 6 · B 6 · C 3 · D 3 · H 0

JUSTIFICATION

Why A* is the right choice

Strategy	Optimal route?	Guided?	Fit for ambulances
Breadth-first	Only if costs equal	No	Ignores travel time
Depth-first	No	No	May wander far away
Uniform-cost	Yes	No	Explores every direction
Greedy best-first	No	Yes	Can pick slow roads
A* search	Yes (admissible h)	Yes	Optimal and focused

A* gives optimal routes while expanding fewer junctions than uniform-cost search.

EXPECTED OUTCOMES

Expected outcomes and real-world impact

Faster response

Shorter time from emergency call to hospital arrival.

Better hospital match

Patients reach a hospital that can actually treat them.

Safer driving

Planned routes mean fewer risky manoeuvres and red-light runs.

Scalable for cities

Built on data feeds cities already have: GPS, traffic and bed counts.

Measured by: response time, time to hospital, route-update rate.

CONCLUSION

Conclusion and future scope

Summary

- Problem: static routing wastes critical minutes
- AI concepts: agent, state space, heuristics, ML
- Strategy: A* with an admissible time heuristic
- Evidence: 68% fewer junctions expanded than Dijkstra

Limits and future scope

- Needs accurate, live traffic data
- Signal priority needs city infrastructure
- Future: reinforcement learning for dispatch
- Future: coordinating many ambulances at once

Thank you · Questions?